

Quantification of Gamma Radiation on Drilling Cuttings Through HPGe Detector: Comparative Study With Well Logs Spectrometry Data

ID:G02-093



D. V. Lamas^{1,2*}, R. C. Florenzano^{1,2}, B. S. N. Rocha¹, A. M. de Souza^{2,3}, L. B. Silva³, J. O. Santos^{2,3}, V. M. J. S. Campos¹, F. Vidal¹, A. X. da Silva³, J. L. Thalhoffer^{1,2,3}, A. F. M. Freire¹

¹Group of Exploratory Interpretation and Reservoir Characterization (GIECAR/UFF);

²Laboratory of Radiobiology and Radiometry (LARARA/UFF);

³Nuclear Engineering Program (PEN/COPPE/UFRJ).

*Correspondence to: lamasdaniel@id.uff.br



1. INTRODUCTION

The Gamma-Ray (GR) log is part of the basic suite of well log tools, key for identifying lithologies and clay content, but spectral gamma-ray tools (eTh232, eU238, K40 measured individually) are costly and not part of standard logging operations. High-Purity Germanium (HPGe) laboratory measurements on drilling cuttings provide a cheaper alternative, motivating this work to compare HPGe results with GR logs, while also integrating X-Ray Fluorescence (XRF) as a reference laboratory tool for major and trace element analysis.

The study was conducted in the interval 4449 m to 4551 m from well 1-BRSA-1007-RJS, located in the south area of the Campos Basin (RJ). These deposits reflect maximum flooding surfaces that mark the transition between shallow and distal marine environments, with prominence given to the Beta Marker, an anomaly recorded in gamma ray and resistivity logs, which marks the contact between the Quissamã formation and the overlying Outeiro Formation (Winter, 2007).

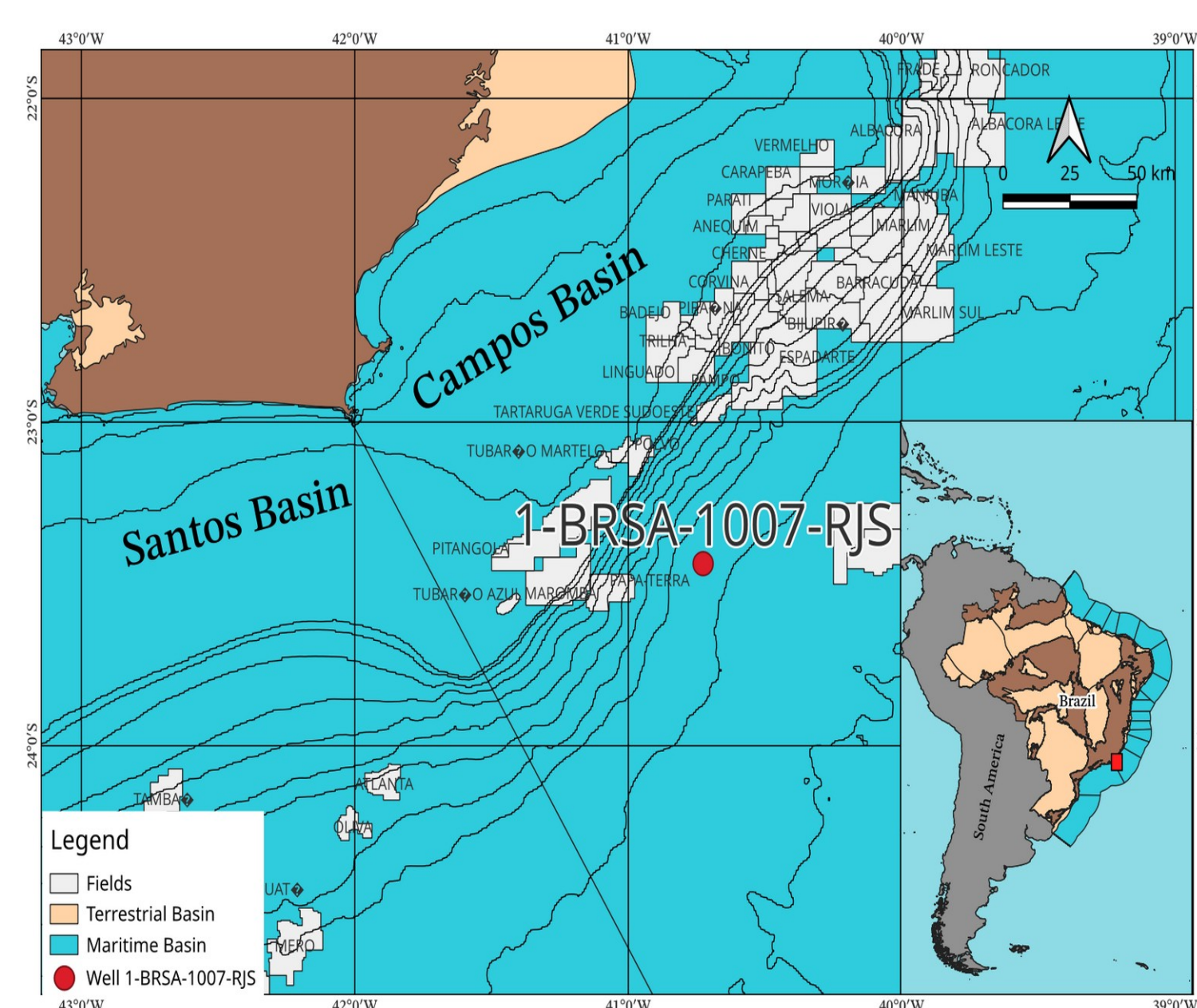


Figure 1: Location of the well studied.

2. MATERIALS AND METHODS

The studied interval is comprised by 35 cutting samples that were separated and processed for measurement on the HPGe and XRF detectors.

Aside from the gamma-ray well log tool, density (RHOB), porosity (NPHI), resistivity (RT) and caliper (CAL) logs were used to monitor the borehole environment as a mean to access gamma-ray attenuation. Their behavior (Figure 3) is well defined and stable, likely reflecting the absence of notable changes in gamma-ray attenuation in the borehole environment.

3. RESULTS

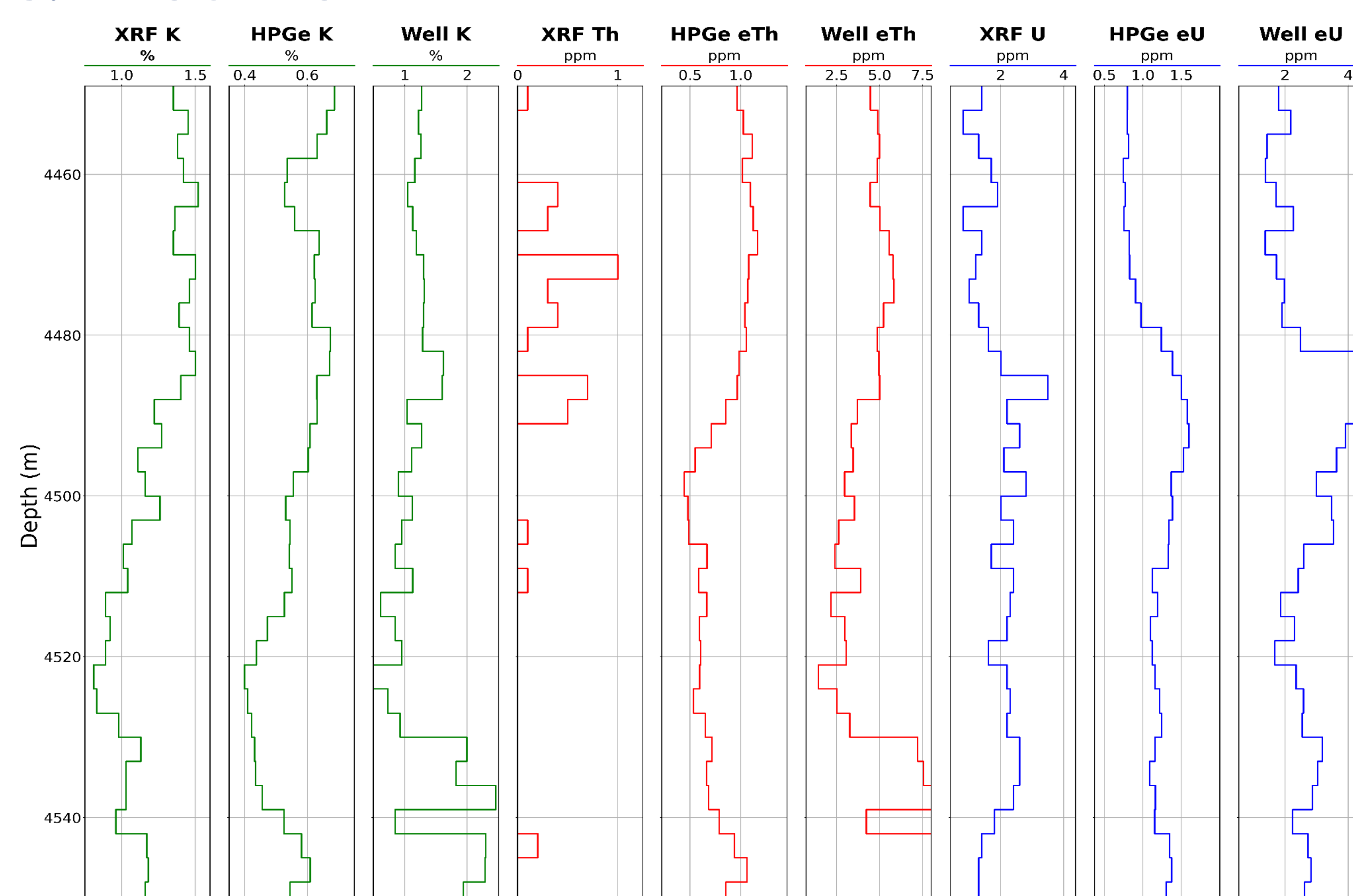


Figure 2: XRF, HPGe and Well logs for K40, Th232 and U238.

It was obtained the isotopic concentration for eTh232 in the photopeaks 238, 338.3, 583, 727.2, 911.6 and 969.1 keV, eU238 in the photopeaks 186.2, 295.2, 351.9, 609.3, 1120 and 1764.5 keV and K40 in the photopeak 1460.8 keV, they were then analyzed qualitatively and quantitatively to access which photopeaks had the best monotonic correlation with the Gamma-Ray well data. Those best photopeaks were then averaged in accordance with their radioelement emission probability (Damascena, 2015), furthermore a moving average was applied to smooth the data.

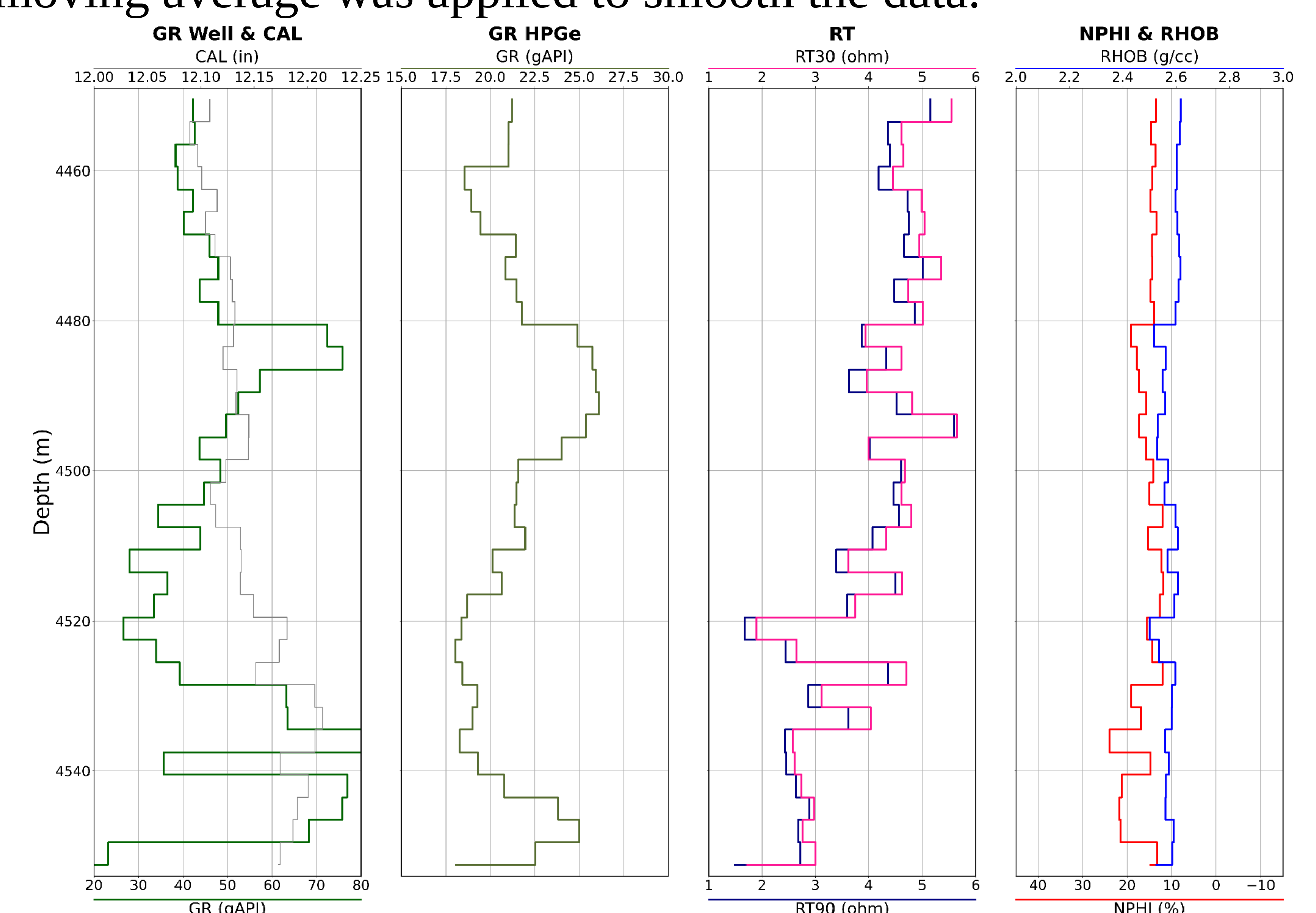


Figure 3: Well GR and CAL, HPGe gAPI GR, RT 30 and RT 90, NPHI and RHOB.

4. CONCLUSION

The radioelement values obtained by HPGe show consistency with the data from the well's gamma-ray tool, as well as the total gAPI gamma-ray calculated through the formula by (Ellis & Singer, 2008). Averaging the results of the best photopeaks of eU238 and eTh232 substantially improved the correlation between the HPGe and well logs, as opposed to analysing each photopeak individually.

Thorium logs only presented meaningful results in laboratory through HPGe measurements, likely because the XRF measurements were close to the detection limit, which is expected in a predominantly carbonate lithology of the Quissamã Formation.

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ACKNOWLEDGEMENTS AND SUPPORT

